# CARBOHYDRATE STRUCTURE CLASSIFICATION FUNCTIONS

# LEARNING OBJECTIVES

By the end of the lecture, the student will be able to :

- Define carbohydrates
- Classify carbohydrates on the basis of functional group and carbohydrate moiety
- Describe the function of mono, di and polysaccharides
- Discuss the biochemical importance of mono, di and polysaccharides

# CARBOHYDRATES

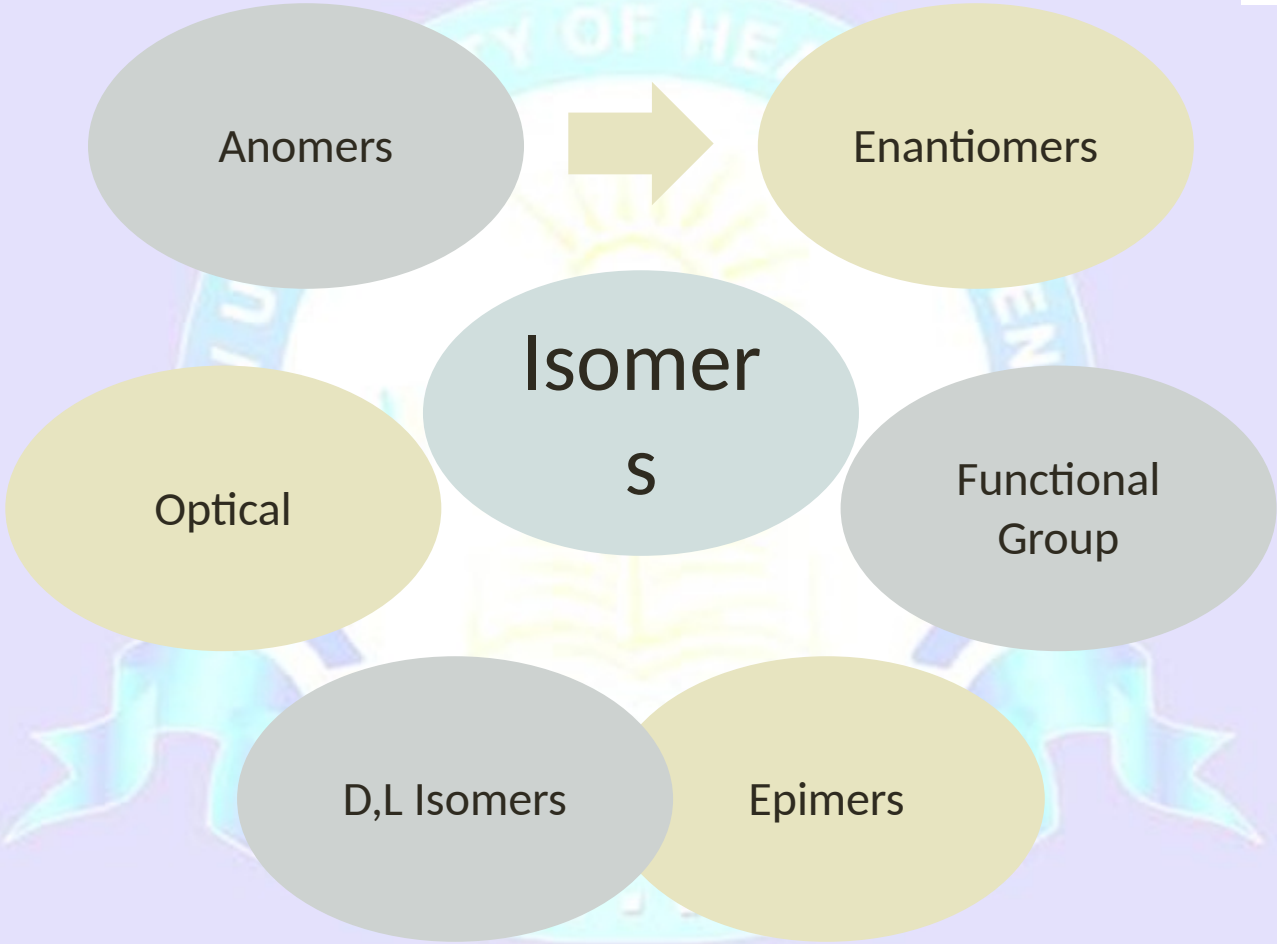
- Any of the group of organic compounds consisting of carbon, hydrogen, and oxygen, usually in the ratio of 1:2:1, hence the general formula: $C_n(H_2O)_n$.
- POLYHYDROXY ALCOHOLS with potentially active ALDEHYDE or KETONE as a functional group
- They serves as a major energy source in the diet of animals.
- includes sugars, starch, cellulose etc.



# CLASSIFICATION

- Divided into four chemical groupings:
- Monosaccharides:
  - Simple sugars with multiple OH groups.
- Disaccharides:
  - 2 monosaccharides are covalently linked.
- Oligosaccharides:
  - 3-10 monosaccharides are covalently linked.
- Polysaccharides:
  - Polymers consisting of chains of monosaccharide or disaccharide units.

# ISOMERS


Anomers
Enantiomers
Isomers
Optical
Functional Group
D,L Isomers
Epimers

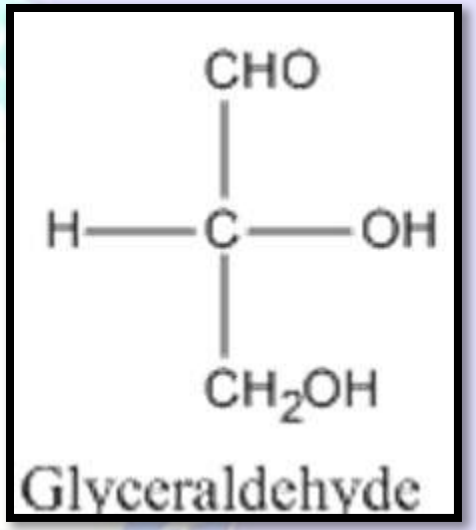
# MONOSSACHARIDES

- The names of most sugars end with the letters ***ose***.
- classified by:
  1. the number of carbons and
  2. whether aldehydes or ketones.
- Based on number of carbons (3, 4, 5, 6), a monosaccharide is a triose, tetrose, pentose or hexose.

# CLASSIFICATION ACCORDING TO FUNCTIONAL GROUP

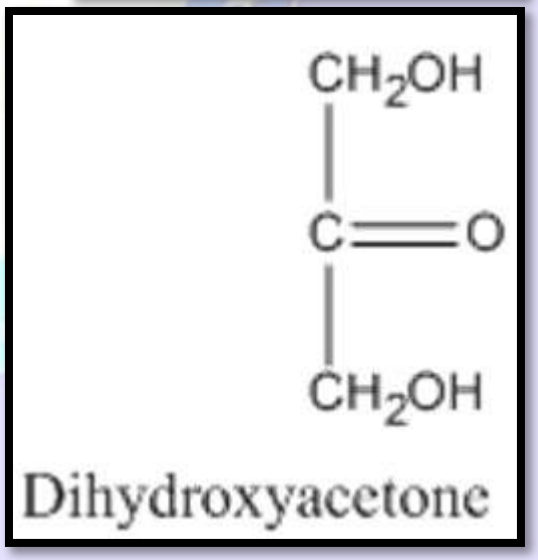
- **Aldehydes:**

  Any of a class of monosaccharide sugars containing an aldehyde group.

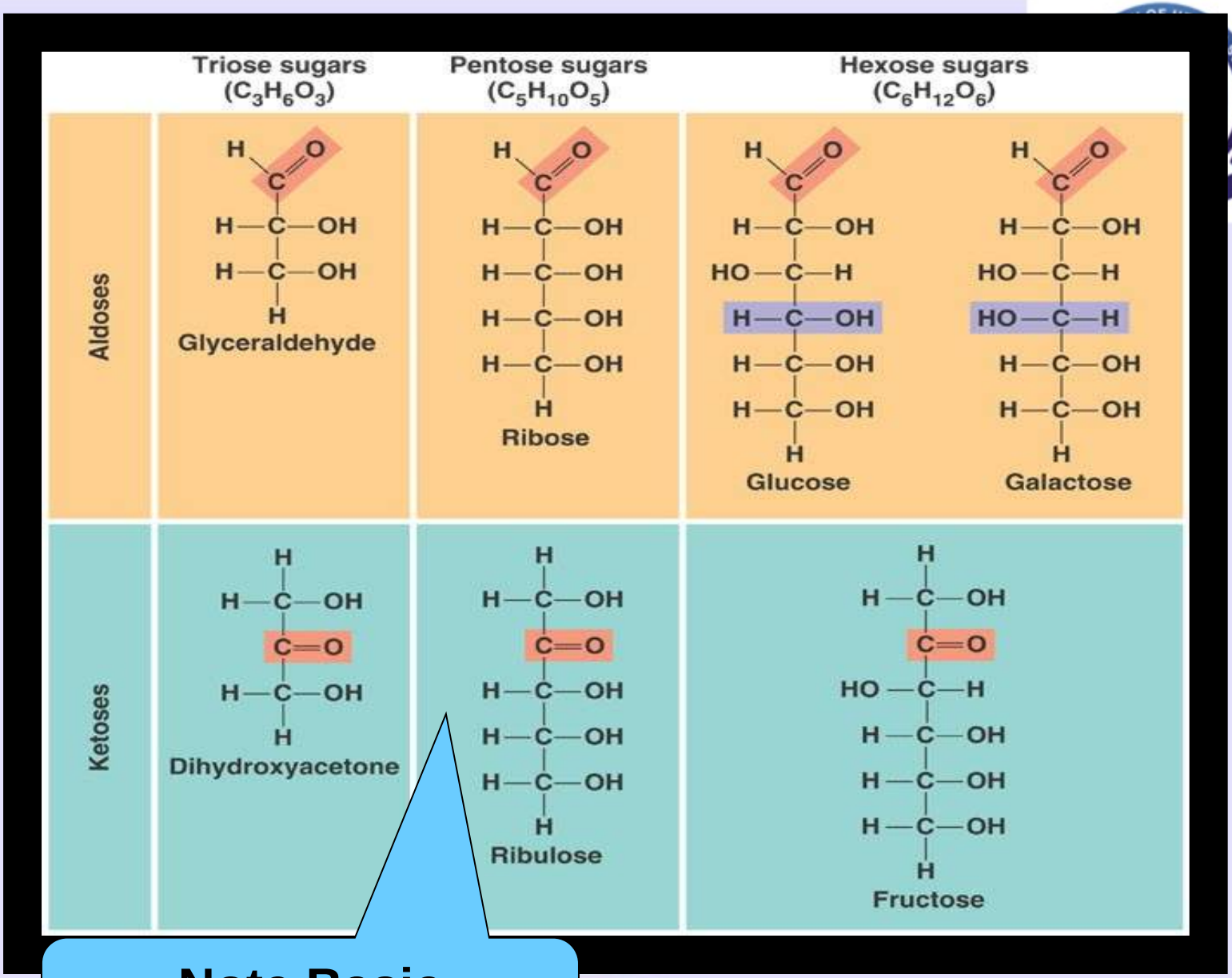



- **Ketones:**

  Any of a class of monosaccharide sugars containing an Ketone group



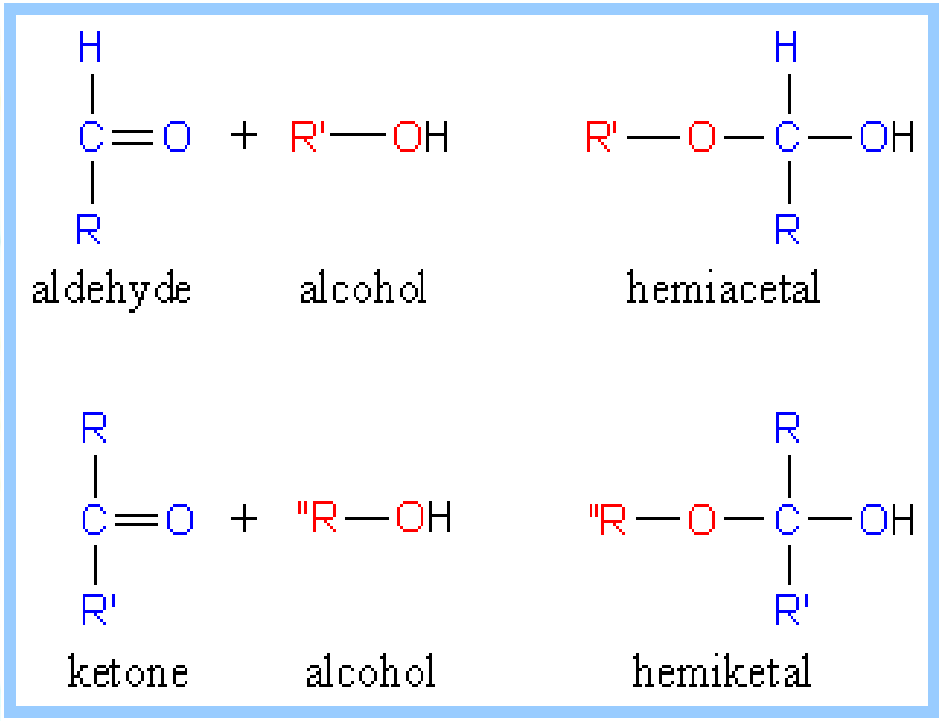
# Some Monosaccharides




**Note Basic Formula: (CH2O)n**

# Hemiacetal / Hemiketal Ring Forms

- An aldehyde can react with an alcohol to form a **hemiacetal**.
- Similarly a ketone can react with an alcohol to form a **hemiketal.**



# Monosaccharide: Functions

| Sugar | Where Found | Biochemical Importance | Clinical Significance |
|---|---|---|---|
| D-Ribose | Nucleic acids. | Structural elements of nucleic acids and coenzymes, eg, ATP, NAD, NADP, flavoproteins. Ribose phosphates are intermediates in pentose phosphate pathway. | |
| D-Ribulose | Formed in metabolic processes. | Ribulose phosphate is an intermediate in pentose phosphate pathway. | |
| D-Arabinose | Gum arabic. Plum and cherry gums. | Constituent of glycoproteins. | |
| D-Xylose | Wood gums, proteoglycans, glycosaminoglycans | Constituent of glycoproteins | |
| D-Lyxose | Heart muscle. | A constituent of a lyxoflavin isolated from human heart muscle. | |
| L-Xylulose. | Intermediate in uronic acid pathway | | Found in urine in essential pentosuria. |

# Monosaccharide: Functions

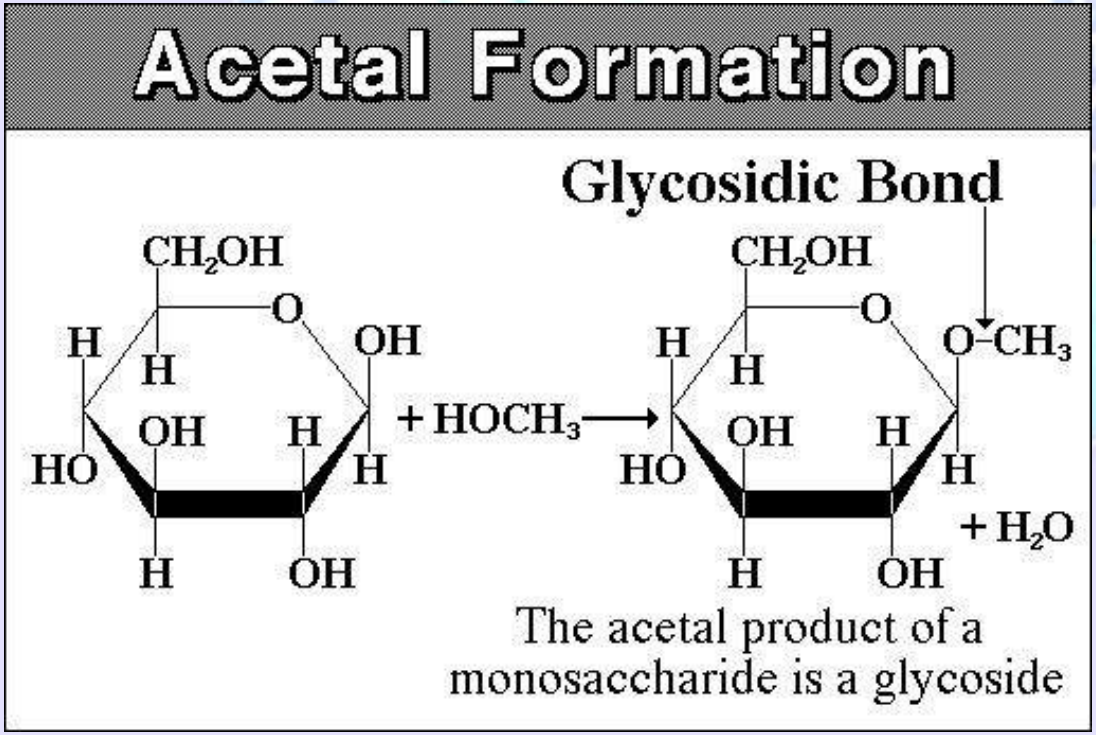
| Sugar | Where Found | Biochemical Importance | Clinical Significance |
|---|---|---|---|
| D-Glucose | Fruit juices. Hydrolysis of starch, cane sugar, maltose, and lactose. | The “sugar” of the body. The sugar carried by the blood, and the principal one used by the tissues. | Present in the urine (glycosuria) in diabetes mellitus owing to raised blood glucose (hyperglycemia) |
| D-Fructose | Fruit juices. Honey. Hydrolysis of cane sugar and of inulin. | Can be changed to glucose in the liver and so used in the body. | Hereditary fructose intolerance leads to fructose accumulation and hypoglycemia. |
| D-Galactose | Hydrolysis of lactose. | Can be changed to glucose in the liver and metabolized. Synthesized in the mammary gland to make the lactose of milk. A constituent of glycolipids and glycoproteins. | Failure to metabolize leads to galactosemia and cataract. |
| D-Mannose | Hydrolysis of plant mannans and gums. | A constituent of many glycoproteins. | |

# FUNCTIONS

- Sources of energy -70-80% human energy needs
- Intermediates in the biosynthesis of other basic biochemical entities (fats and proteins)
- Associated with other entities such as glycosides, glycoproteins, Sialic acids, vitamins and antibiotics)
- Form structural tissues in plants and in microorganisms (cellulose, lignin, murein)
- Participate in biological transport, cell-cell recognition, activation of growth factors, modulation of the immune system
- Formation of DNA and RNA

# DISACCHARIDES

- Disaccharides are composed of 2 monosaccharides joined together by a glycosidic bond.
- Glycosididc Bond
  - O and N



# Disaccharides

| Sugar | Source | Clinical Significancce |
|---|---|---|
| Maltose | Digestion by amylase or hydrolysis of starch. Germinating cereals and malt. | |
| Lactose | Milk. May occur in urine during pregnancy. | In lactase deficiency, malabsorption leads to diarrhea and flatulence. |
| Sucrose | Cane and beet sugar. Pineapple. Carrot roots. | In sucrase deficiency, malabsorption leads to diarrhea and flatulence. |

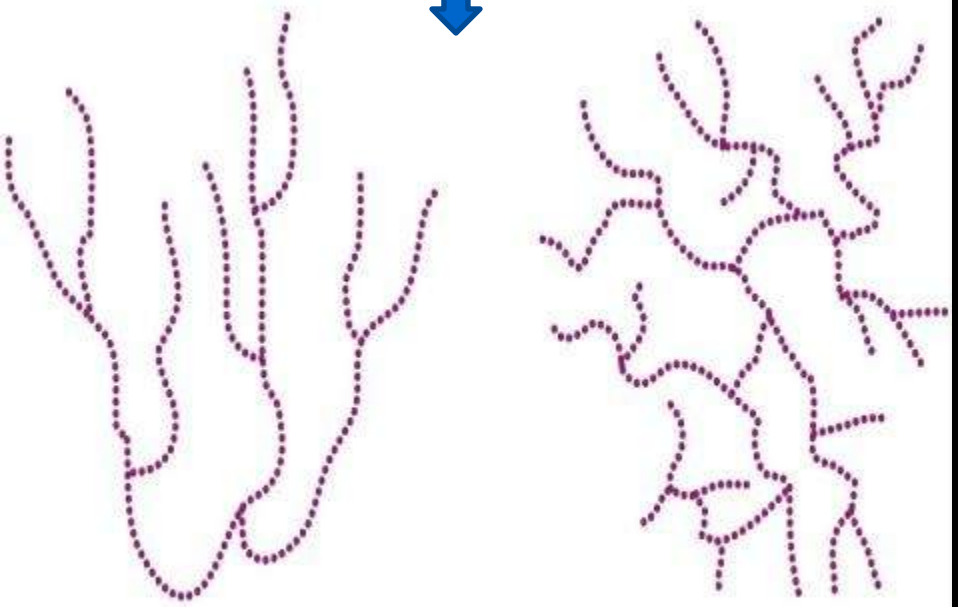
# POLYSACCHARIDES

- Monosaccharides are bonded together to form long chains called polysaccharides.
- Classified as
  - Homopolysaccharides
    - Same monomers
    - Animals store extra carbohydrates as glycogen in the liver and muscles.
    - Stored Polysaccharides serve as the immediate source of energy
  - Heteropolysaccharides
    - Different monomers
    - Non carbohydrate moiety
    - They form proteoglycans

# POLYSACCHARIDES

- Starch and glycogen are polysaccharides that function to store energy.
- They are composed of glucose monomers bonded together producing long chains.
- Starch is less while glycogen is highly branched structure
- Cellulose is non-digestable plant sugar

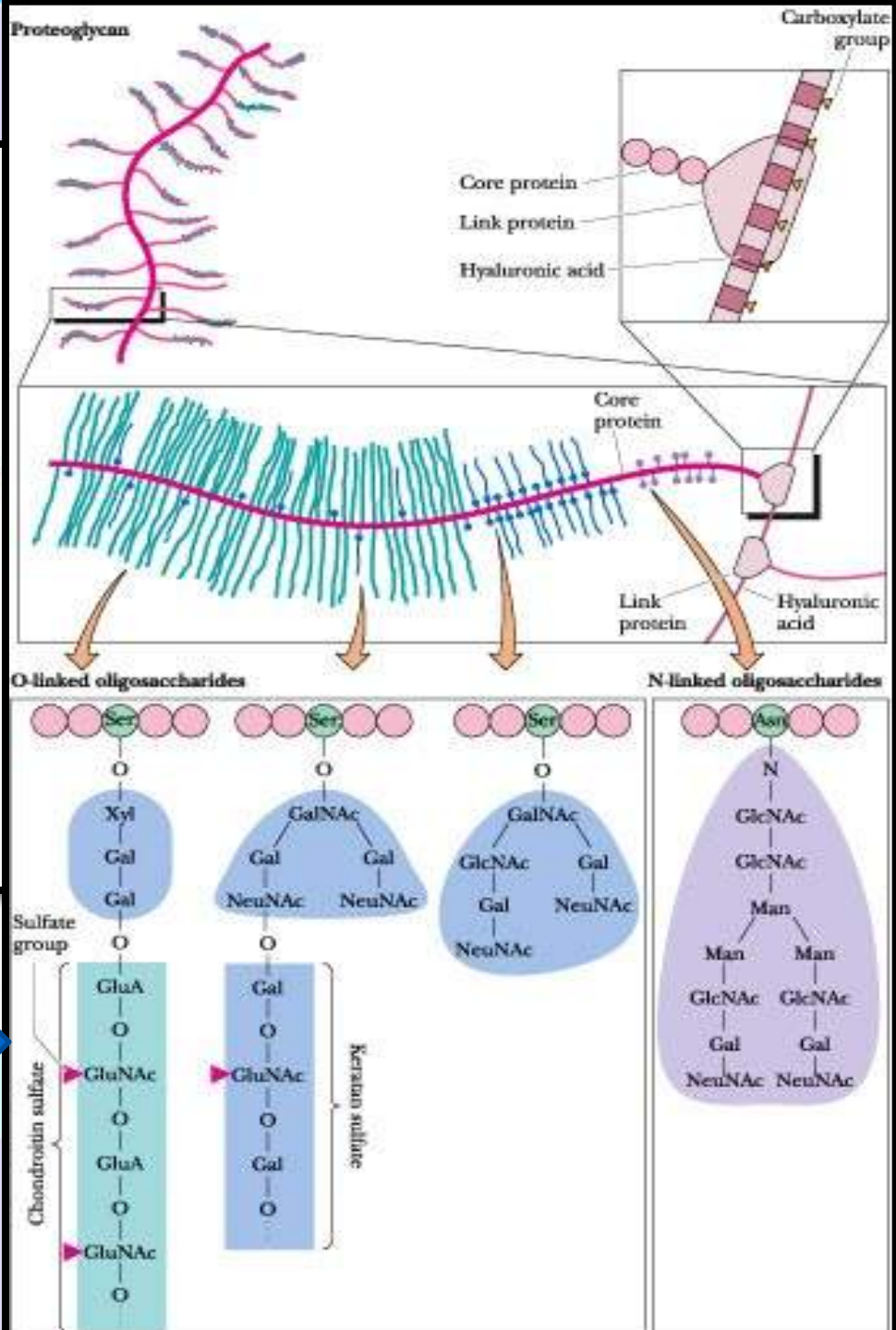
# Homopolysaccharide



Amylopectin
(in plants)

Glycogen
(in animals)

# Heteropolysaccharide



# LACTOSE INTOLERANCE

- Infants who are lactase – deficient are less able to metabolize lactose.
- Treatment is to reduce consumption of milk while eating yogurts and cheeses, as well as green vegetables .
- Use lactase treated products.



# References

- Harper chapter “carbohydrates of physiologic significance”
- Pages113-120

# THANK YOU